CO1 - AT3

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(1A) Intelligent agents in a smart home system,

An Intelligent agents Perceives its environment through sensors and acts on it through actuators to achieve its objectives,

Types of agents:

1, Simple reflex agent:

- How it works:-

It uses condition - action rules on the current percept only. on memory or model of the world,

In the smart home:-

- If motion is dected and it is dark, turn on the light,

Strength :- fast, simple, and reliable, for security,

limitation :- It cannot learn Preferences,

2, model- based reflex agent:-

It maintains an internal state a model of how, world evolves and how its action,

→ Strength :- It handles incomplete information and tracks history,

→ limitations :- It still reacts by rules and has no of goals or of how good an outcome in,

3. Goal - based agent :-

It has explicit goals and chooses actions by searching or planning toward them,

→ A Room at 24°C by 6 PM when the user returns. the agent plane to 100k doors,

strength :- It is flexiable. because changing the goal changes the behaviour without recoriting rules,

limitations :- It cannot distinguish between multiple ways of reaching a goal,

4, utility - based agent :-

It uses a utility function that assigns a numeric value to each state,

Comfort :- keep the temperature near the user's preferred value,

Security :- weigh the risk of a false alarm against a missed instrusion,

strength :- It handles trade - off and gives graded,

limitation :- It is computationally heaviour and needs a good a utility model.

=) Justification :- a hybrid (layered) approach :-

No single agent type is best, the most effective design is a hybrid, layered architecture,

Comparison :-

Agent	memory	Goals	Trade-offs	Smart home
Simple	NO	NO	NO	Instant triggers (alarm, motion, light)
model based	Yes	NO	NO	occupancy,
Goal - based	Yes	Yes	NO	scheduled
Utility based	Yes	Yes	Yes	Balancing comfort,

Conclusion:- A comfort (Preferences learned and Respected). energy efficiency is verified a no single agent type can deliver alone,

(2A) Robot in an unknown maze,

Problem formulation :-

State :- robot's Position

Actions :- up, down, left, right

Goal :- reaches the exit.

Path cost :- cost per move.

No heuristic available :- so, use uninformed search,

Uniform cost search (UCS):

- Expands the lowest path cost $g(n)$ first (Priority queue).
- Complete if step cost $\geq \varepsilon > 0$; optimal
- Time and space : $O(b^{1+\lfloor C^*/\varepsilon \rfloor})$
- Advantages :- guarantees the cheapest path :- handles varying costs.
- Disadvantages :- very high memory, explores in directio

Iterative Deepning search (IDS):

- Repeats depth-limited DFS with limits 0, 1, 2..
- Complete ; optimal for uniform step costs.
- Time : $O(b^d)$; space : $O(bd)$
- Advantages : low memory ; no need to know depth in advance.

Complexity :-

b = branching factors, d = depth of shallowest goal, C^* = optimal cost, ε = minimum step cost,

	Time	Space	Advantages.
UCS	$O(b^{1+\lfloor C^*/\varepsilon \rfloor})$	$O(b^{1+\lfloor C^*/\varepsilon d \rfloor})$	optimal for any positive costs
IDS	$O(b^d)$	$O(b \cdot d)$	linear memory. complete, optimal for equal costs,

In a maze b is small, so IDS's repeated work is minor, which UCS's memory can run not large maze.

	Behavior in the maze	Effect.
reflex	follows a rule such as "turn left"	loops forever in mazes with cycles, no memory.
-based	stores a map of visited cells.	Avoids loops and cores with Partial observability
-based	searches for a Path to the exit	Plans instead of wandering.
ility based	weights Path. length. and risk.	Best when moves have different costs.

The agent's design decides what it remembers and what it optimizes.
The search strategy decides how it explores.

→ Varying costs ⇒ A utility-based agent using UCS, which guarantees the cheapest Path at the price of more memory.

Conclusion:
Pair the agent Type with the search strategy A model based goal agent with IDS is the Practical default.